L

LV.0 → LV.1 · AI ENGINEER

11 L-words every Level 0 → 1 AI engineer must know.

LLMs, LoRA, LSTM, Loss Function & more. **Plain-English** definitions, one swipe each, plus the L-named tools and minds to follow.

↓ UP NEXT · TERMS

The L stack.

LLMs · LoRA · LSTM · Loss Function · Learning Rate · Lemmatization · Linear Regression · Logistic Regression · Label Encoding · LR Scheduler · Layer Normalization

L

LEMMATIZATION · NLP

Lemmatization

strip words to their root meaning

Lemmatization reduces words to their base dictionary form using context and grammar - so "running", "ran", and "runs" all become **"run"**, improving text understanding beyond naive stemming.

USED IN

- NLP preprocessing
- Search
- Text classification

WORD FORMS TO BASE LEMMA

LEARNING RATE · TRAINING

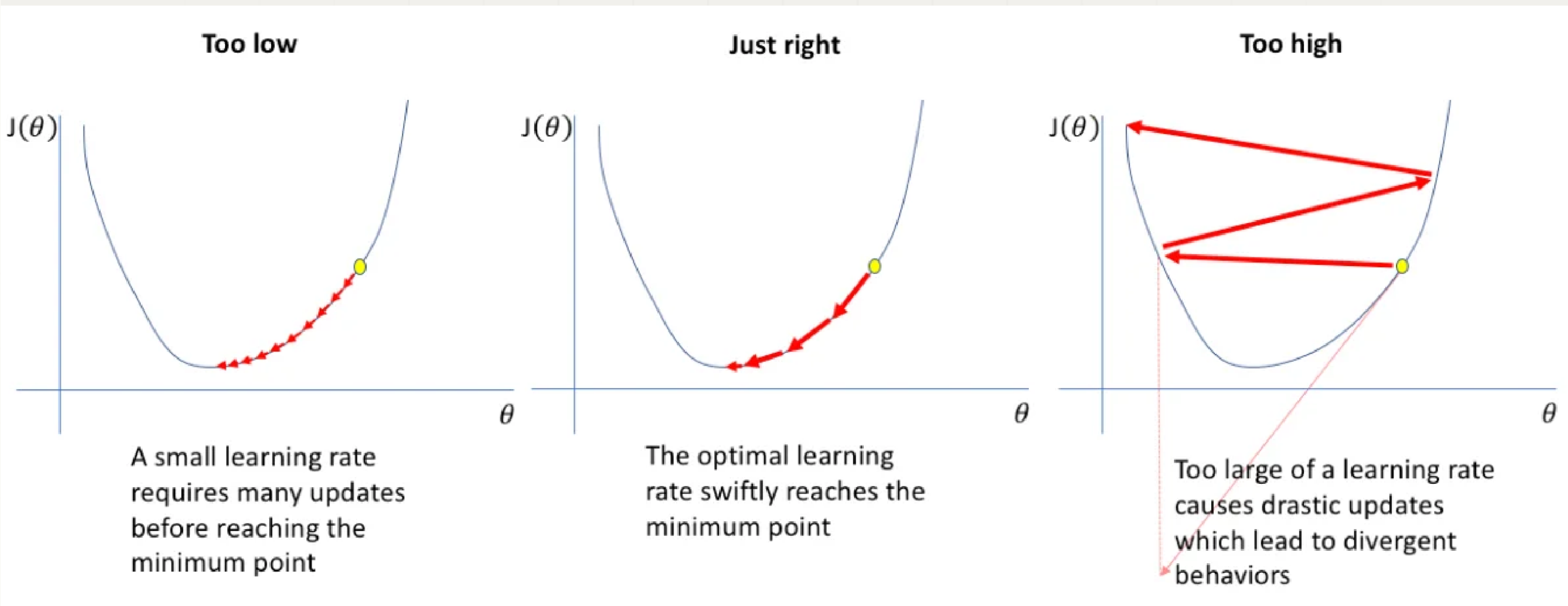
Learning Rate

how big each training step is

The learning rate controls how much model weights shift per gradient step. **Too high and training explodes. Too low and it crawls.** Getting it right is one of the most impactful choices in ML.

USED IN

- Model training
- Hyperparameter tuning
- Optimization



STEP SIZE ALONG LOSS LANDSCAPE

LONG SHORT-TERM MEMORY · ARCHITECTURE

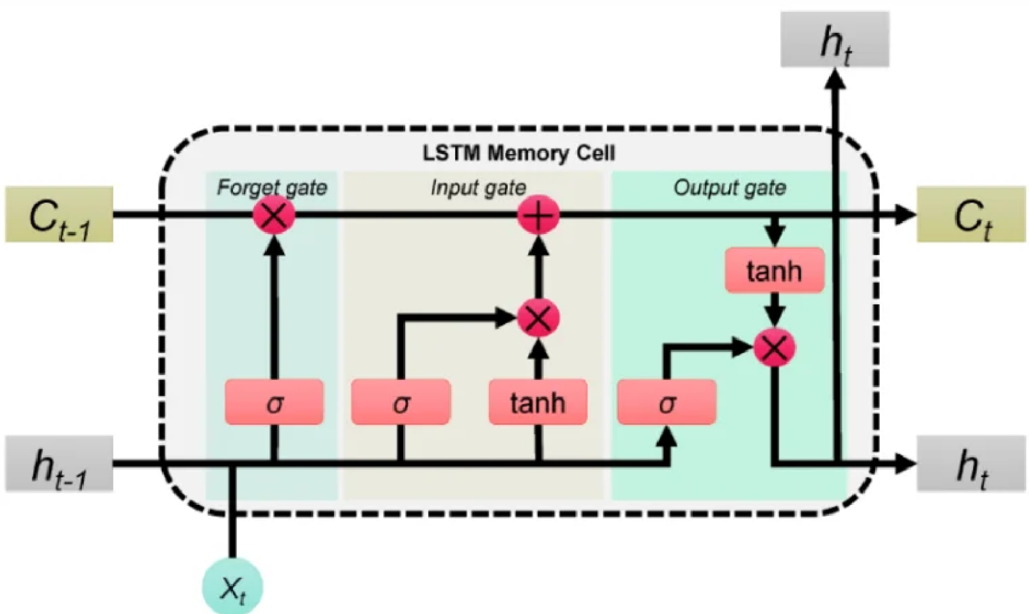
LSTM

remember the past, forget the noise

LSTMs use memory cells and gates to **selectively retain information across long sequences** - the architecture that powered speech and translation before transformers.

USED IN

- Sequential data
- Speech recognition
- Time-series



INPUT, FORGET, OUTPUT GATES

LINEAR REGRESSION · LEARNING

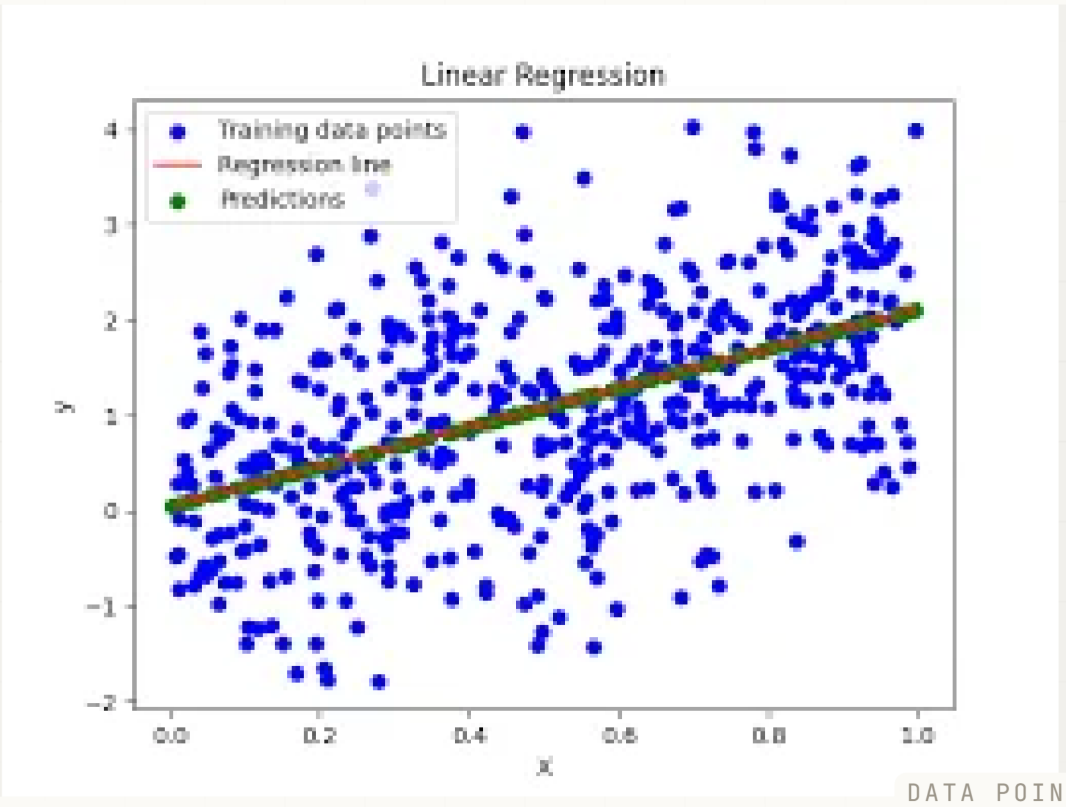
Linear Regression

the best-fit line through data

Linear regression fits a straight line through data to **predict a continuous output from one or more inputs** - the most fundamental algorithm in predictive analytics and ML.

USED IN

- Predictive analytics
- Forecasting
- Feature analysis



DATA POINTS, BEST-FIT LINE

LOW-RANK ADAPTATION · FINE-TUNING

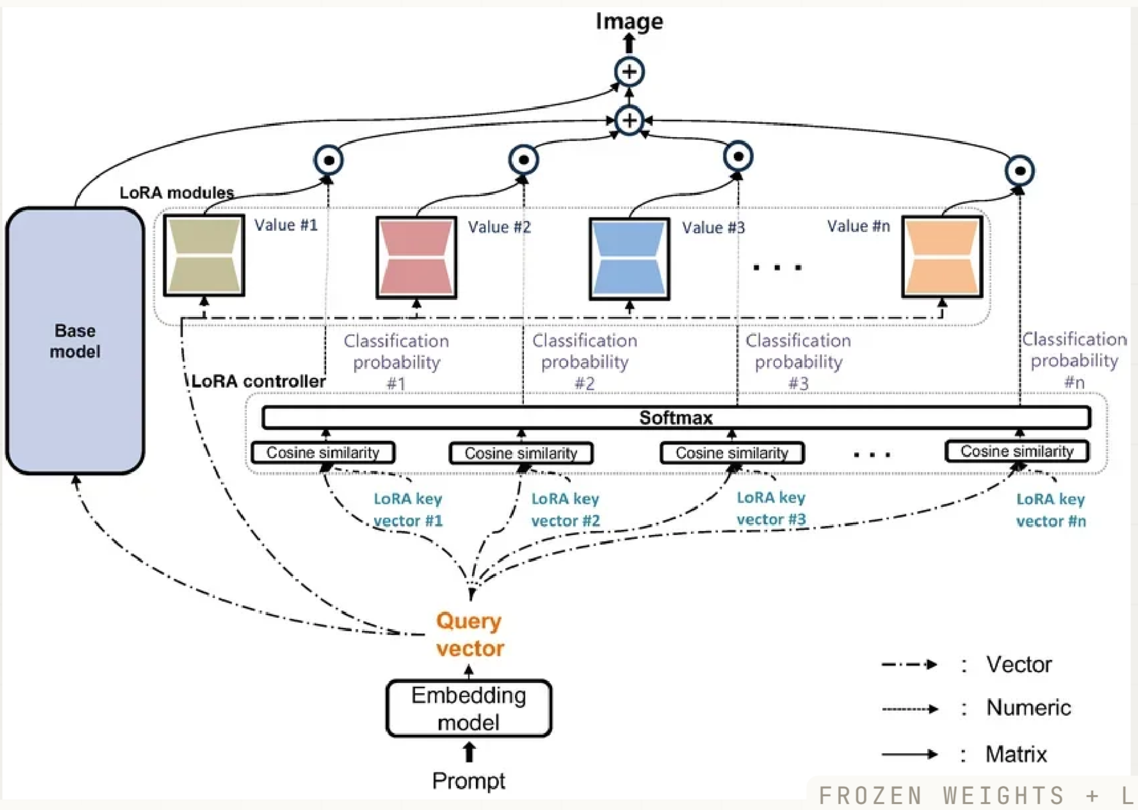
LoRA

tune the delta, freeze the base

LoRA fine-tunes LLMs by training only a tiny set of low-rank weight matrices added on top of the frozen base model - **cutting memory by 10x while matching full fine-tune quality**.

USED IN

- LLM fine-tuning
- PEFT
- Custom adapters



FROZEN WEIGHTS + LOW-RANK ADAPTER

LOGISTIC REGRESSION · LEARNING

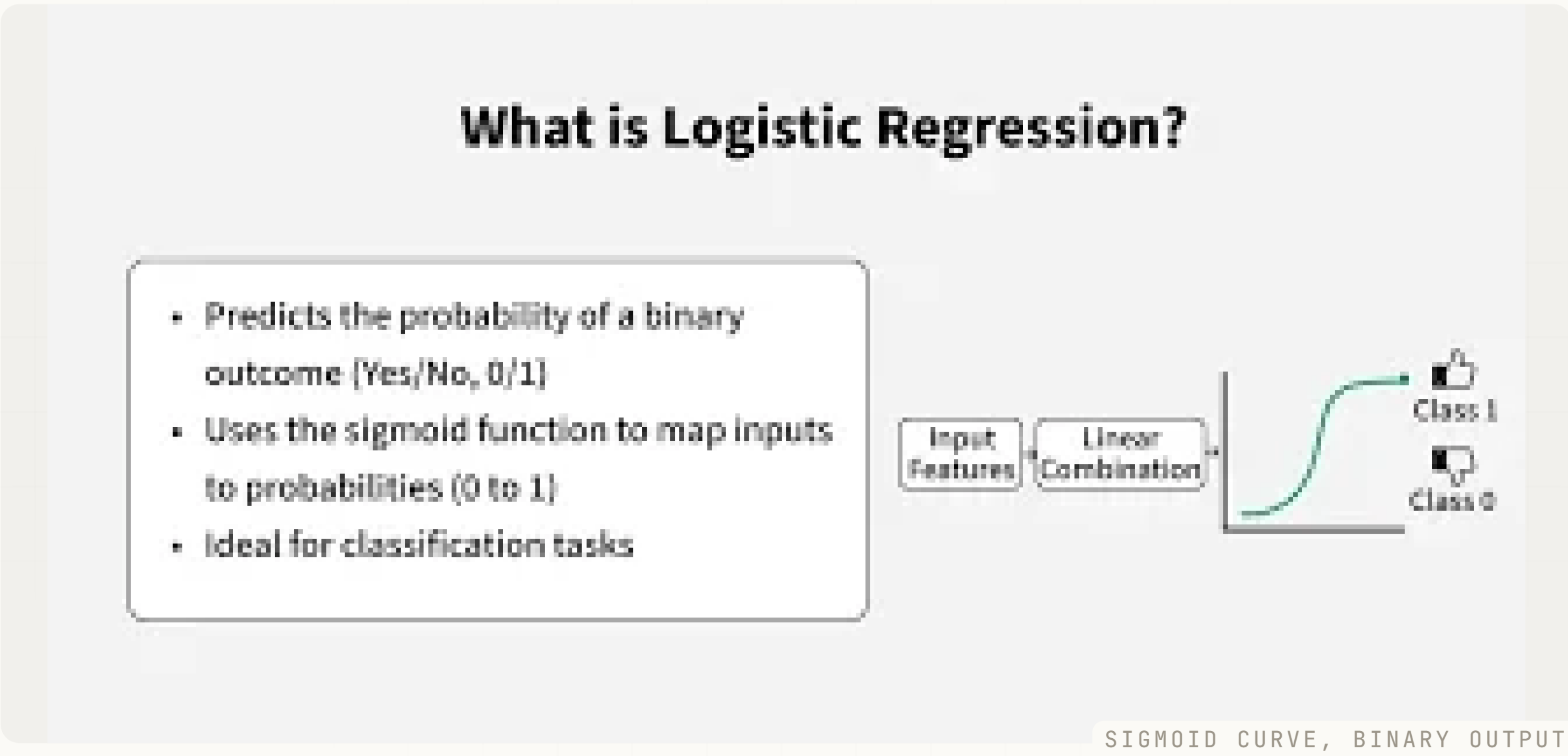
Logistic Regression

predicts probability, not a number

Despite the name, logistic regression is a **classification algorithm** - it predicts the probability of belonging to a class, powering fraud detection, medical diagnosis, and customer analytics.

USED IN

- Classification
- Fraud detection
- Healthcare ML



SIGMOID CURVE, BINARY OUTPUT

LARGE LANGUAGE MODELS · MODELS

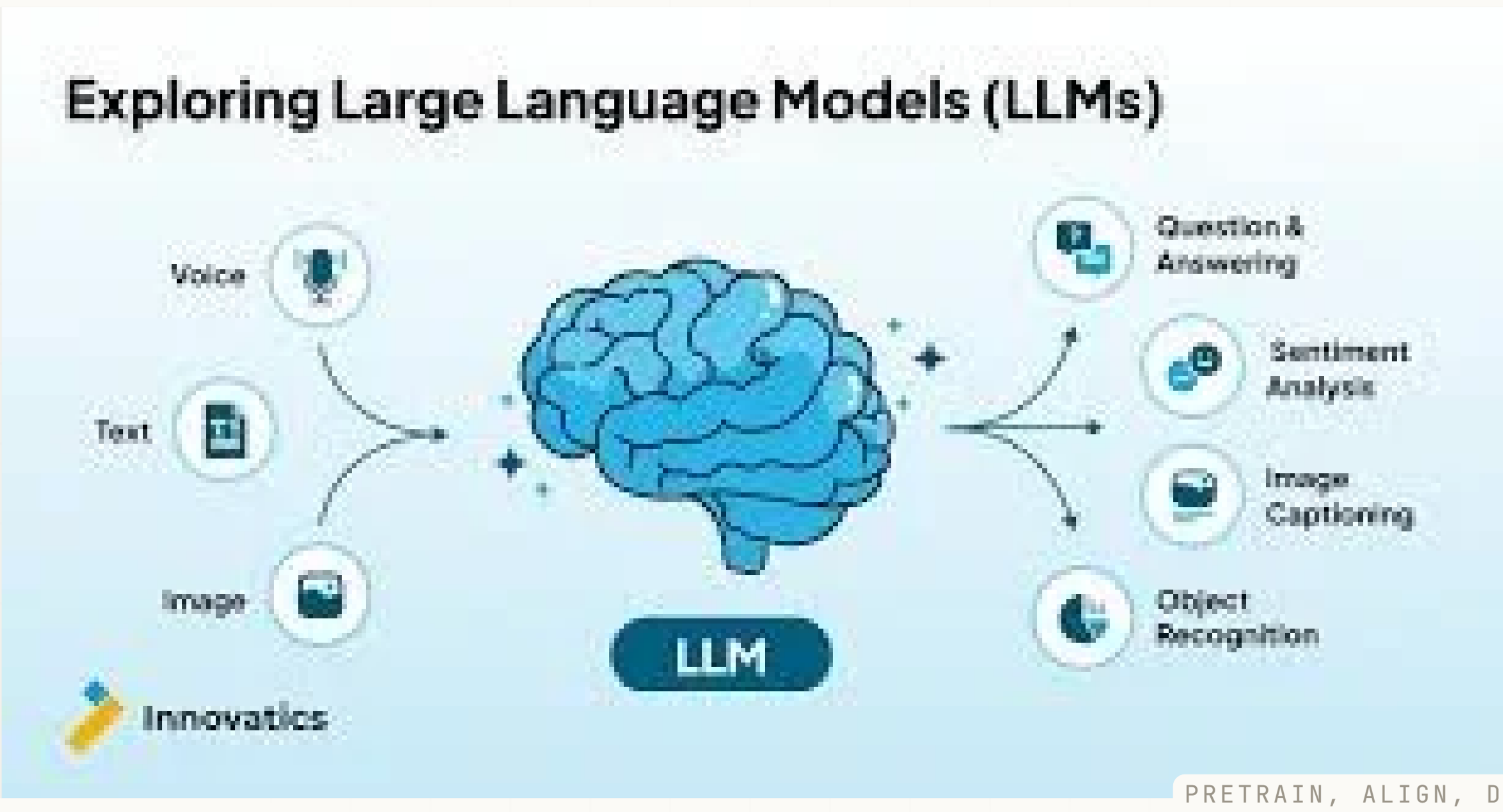
LLMs

the engine behind modern AI

LLMs are trained on massive text corpora to **understand and generate language**, powering assistants, coding tools, agents, search, and nearly every AI application being built today.

USED IN

- AI assistants
- Coding tools
- Agentic systems



PRETRAIN, ALIGN, DEPLOY

LOSS FUNCTION · TRAINING

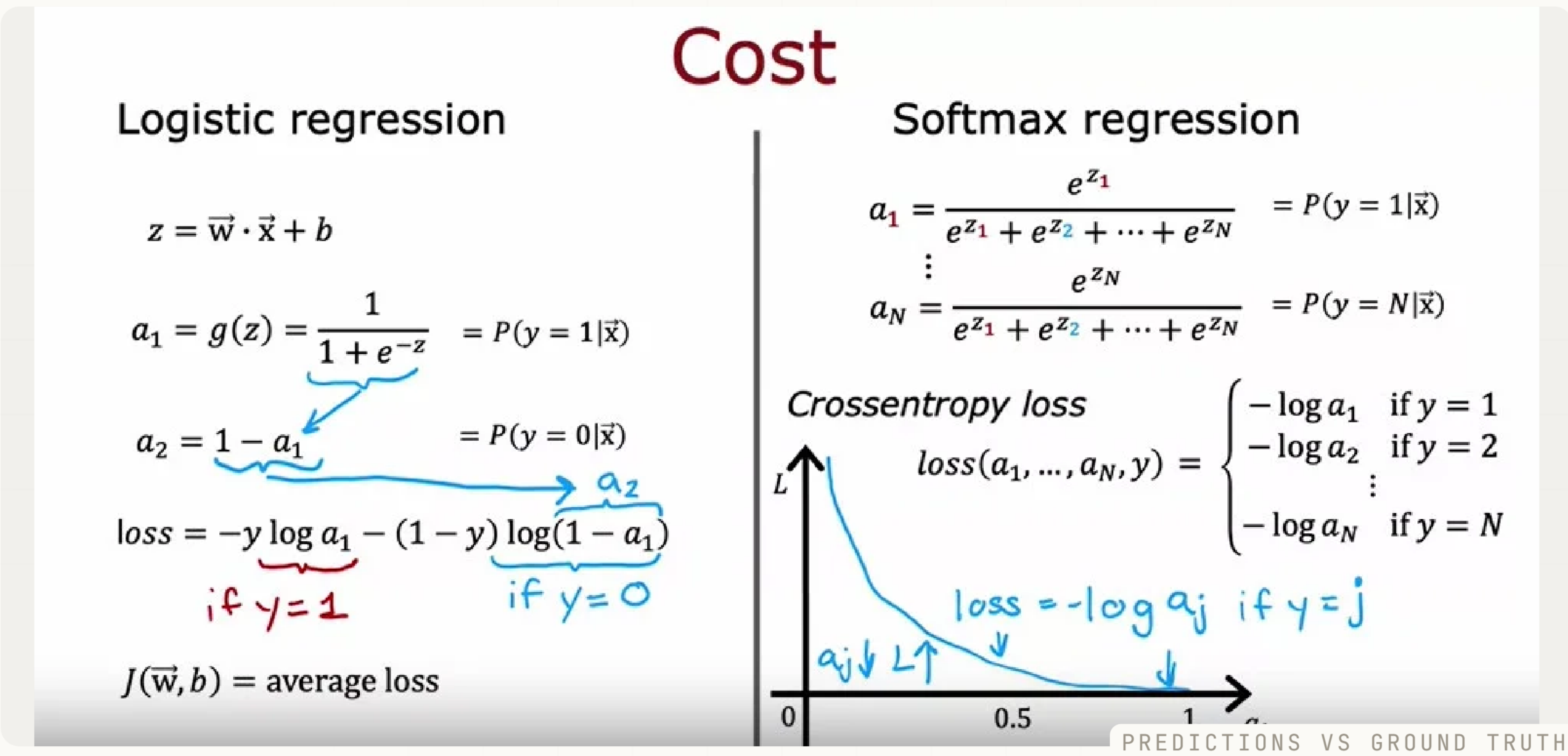
Loss Function

how wrong is the model right now

The loss function measures the gap between predictions and ground truth. **Training is the process of minimizing this number** - everything from backprop to optimizers exists to reduce it.

USED IN

- All ML training
- Backpropagation
- Model evaluation



PREDICTIONS VS GROUND TRUTH

LABEL ENCODING · DATA

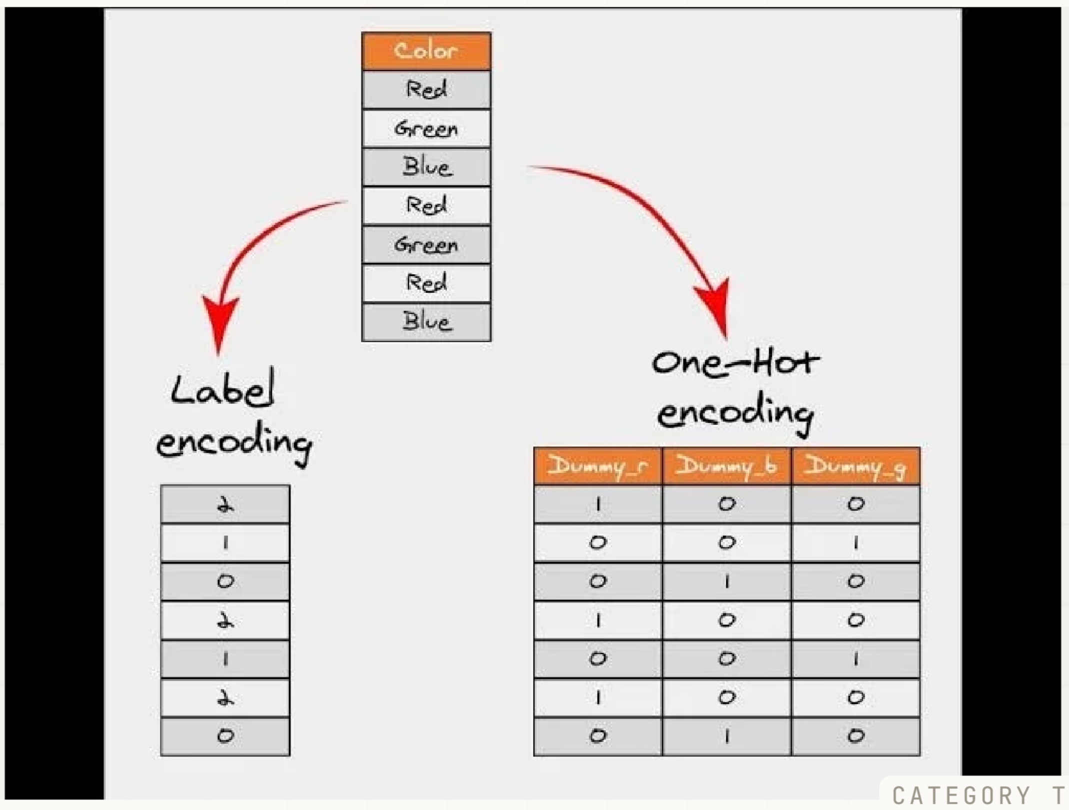
Label Encoding

categories in, numbers out

Label encoding converts categorical text values into integers so ML models can process them. **A required preprocessing step** before training on any non-numeric feature like color, city, or product type.

USED IN

- Feature engineering
- Data preprocessing
- Classification



CATEGORY TO INTEGER MAPPING

LR SCHEDULER · TRAINING

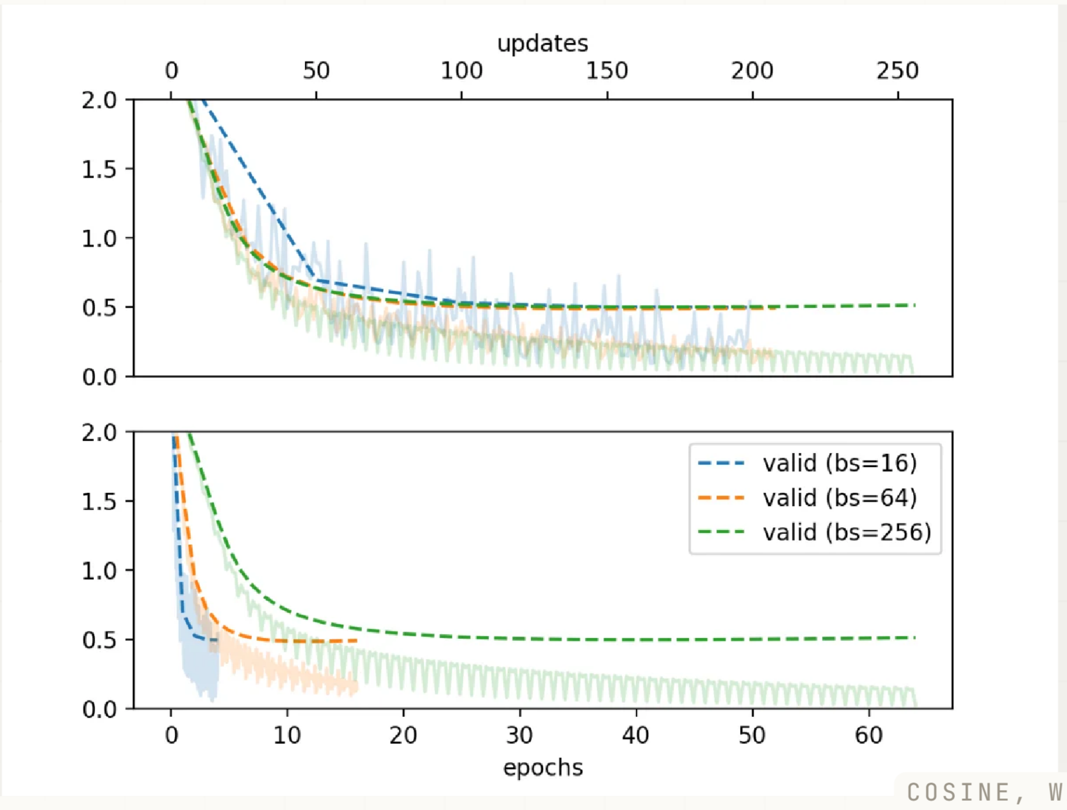
Learning Rate Scheduler

start fast, finish precise

An LR scheduler dynamically changes the learning rate during training - often **large early to move fast, then decaying to fine-tune**, leading to better convergence and final model quality.

USED IN

- Deep learning
- LLM training
- Convergence tuning



COSINE, WARMUP, STEP DECAY

LAYER NORMALIZATION · ARCHITECTURE

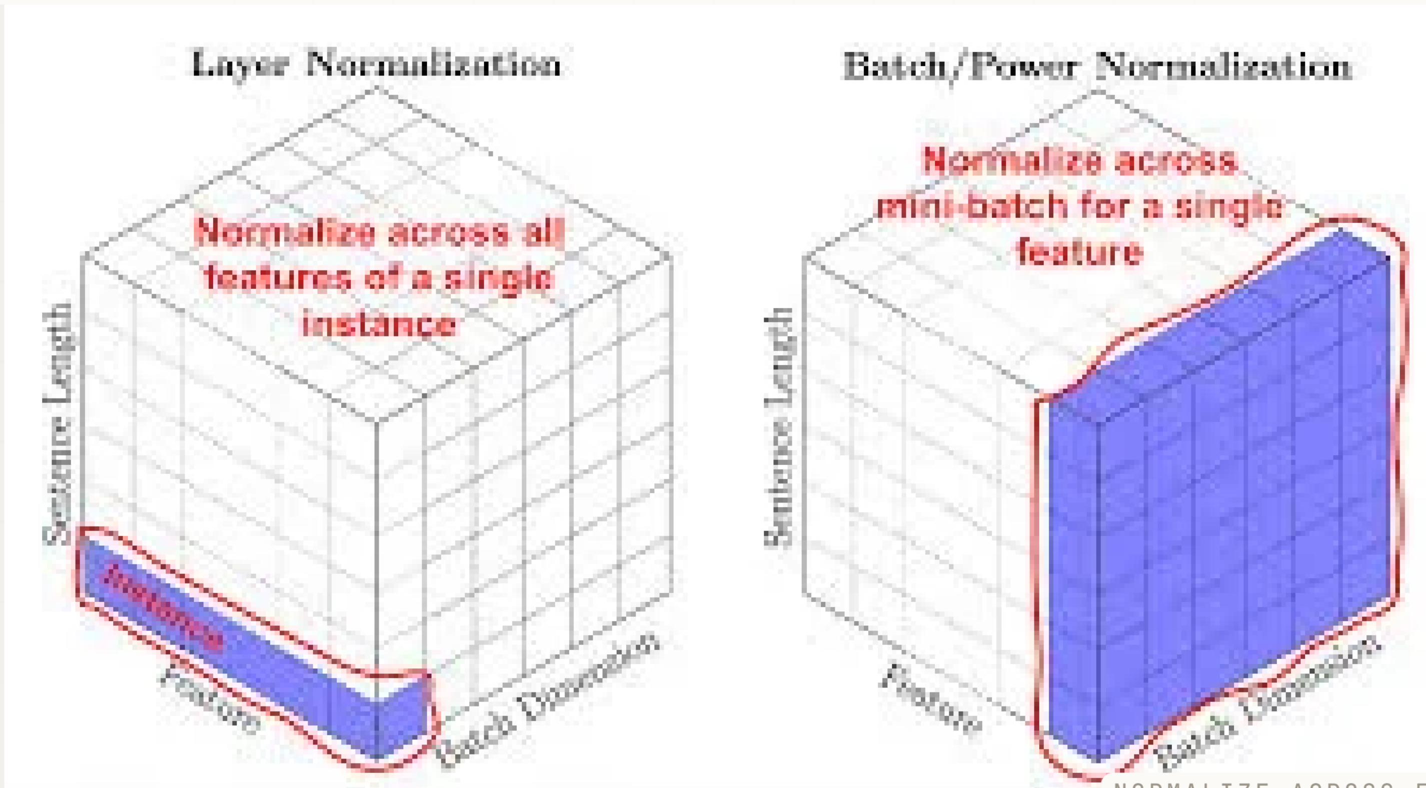
Layer Normalization

stabilize activations every layer

Layer norm standardizes activations within each layer. **Every transformer uses it** to prevent instability and enable deeper, faster-converging networks.

USED IN

- Transformers
- LLMs
- Deep networks



NORMALIZE ACROSS FEATURES

↓ UP NEXT · COMPANIES & TOOLS

The L labs.

Six L-named tools and platforms: **LangChain, LlamaIndex, Llama**, and the infrastructure every AI engineer reaches for first.

L

LLM APP FRAMEWORK

LangChain

prompts · retrieval · agents · chains

The most widely adopted framework for building LLM-powered apps - **wiring together prompts, memory, retrieval, tools, and agents** into production workflows with minimal boilerplate.

2022
Founded by Harrison Chase

90K+
GitHub stars

Agents
Most-used agentic AI framework

WHAT IT DOES

Orchestrates LLMs, tools, memory, and retrieval.

WHY IT MATTERS

De facto standard for LLM app development.

BEST FOR

RAG pipelines, agents, and chatbots.

FIND THEM

langchain.com · github.com/langchain-ai

DATA FRAMEWORK FOR LLMS

LlamaIndex

connect LLMs to your data

LlamaIndex connects LLMs to external data - documents, databases, and APIs - making it the go-to framework for RAG pipelines, knowledge assistants, and intelligent enterprise search.

2022
Founded by Jerry Liu

RAG
Primary use: retrieval-augmented gen

30K+
GitHub stars

WHAT IT DOES

Indexes and retrieves data for LLMs.

WHY IT MATTERS

The standard data layer for RAG apps.

BEST FOR

Enterprise search and doc Q&A.

FIND THEM

llamaindex.ai · github.com/run-llama

LLM BENCHMARKING PLATFORM

LM Arena

human preference · head-to-head · open

A community-driven platform where humans **vote on which LLM response is better** in blind head-to-head comparisons - the most influential real-world model leaderboard in AI.

ELO
Chess-style ranking system

Human
Preference votes, not benchmarks

Open
Anyone can vote and compare

WHAT IT DOES

Ranks LLMs by human preference votes.

WHY IT MATTERS

The most trusted public model leaderboard.

BACKED BY

UC Berkeley LMSYS research group.

FIND THEM

lmarena.ai

LOCAL LLM DESKTOP APP

LM Studio

run LLMs offline · on your laptop

A desktop app for running open-source LLMs locally with no cloud, no API keys, and no cost - **download, manage, and chat with any model** directly on your Mac, Windows, or Linux machine.

Local
100% offline, no API key needed

Free
Free for personal use

Any model
Llama, Mistral, Qwen, and more

WHAT IT DOES

Downloads and runs LLMs on your laptop.

WHY IT MATTERS

Private, free, offline AI for developers.

BEST FOR

Testing models without cloud costs.

FIND THEM

lmstudio.ai

AI DATA PLATFORM

Labelbox

label · evaluate · improve

An AI data platform for **labeling training data, evaluating model outputs, and improving AI quality** - the infrastructure layer teams use to build reliable ML and generative AI systems.

2018
Founded in San Francisco

Data
Labeling, curation, and evals

RLHF
Powers human feedback pipelines

WHAT IT DOES

Data labeling and AI quality assurance.

WHY IT MATTERS

Quality training data drives model quality.

BEST FOR

Enterprise ML teams and GenAI pipelines.

FIND THEM

labelbox.com

OPEN-WEIGHT MODEL FAMILY

Llama

Meta · open weights · frontier

Meta's open-weight LLM family - **the most influential open-model release in AI history**, enabling millions of developers, researchers, and companies to build, fine-tune, and deploy frontier AI.

Meta

Developed by Meta AI Research

Open

Open weights for research and commercial use

Llama 3

Latest generation, frontier quality

WHAT IT IS

Meta's open-weight frontier LLM series.

WHY IT MATTERS

Made frontier AI accessible to everyone.

USED FOR

Fine-tuning, research, and deployment.

FIND THEM

ai.meta.com/llama · huggingface.co/meta-llama

↓ UP NEXT · CHANNEL TO FOLLOW

Learn from the best.

One L-channel worth bookmarking: the **podcast and newsletter** that covers the AI engineer ecosystem better than anyone else.

L

AI MEDIA · PODCAST · NEWSLETTER

Latent Space

@latentspacepod · latent.space

The leading AI engineer podcast and newsletter covering **LLMs, open-source AI, startups, and emerging research** - the highest signal source for developers following the AI ecosystem.

Podcast

Deep interviews with AI builders

Newsletter

Weekly AI research and product news

Engineers

Made by and for AI engineers

BEST FOR

AI engineers keeping up with the field.

COVERAGE

LLMs, open-source, startups, research.

FORMAT

Podcast, newsletter, and YouTube.

FIND THEM

latent.space · @latentspacepod

↓ UP NEXT · PEOPLE TO FOLLOW

The L minds.

Three L-named educators and researchers worth a follow - making **deep learning and AI** accessible to millions worldwide.

L

AI EDUCATOR · GOOGLE

Laurence Moroney

TensorFlow · Coursera · @lmoroney

Google's former AI Advocate and the educator who made **TensorFlow and practical deep learning** accessible to millions of developers through books, courses, and hands-on content.

Google
Former AI Advocate at Google

TF
TensorFlow educator and author

Coursera
Millions of learners through courses

KNOWN FOR

TensorFlow tutorials and ML courses.

WHY FOLLOW

Practical deep learning for developers.

BEST FOR

Hands-on TF and applied ML skills.

FIND THEM

laurencemoroney.com · linkedin.com/in/laurence-moroney

AI EDUCATOR · SERRANO ACADEMY

Luis Serrano

Serrano Academy · @SerranoAcademy

A mathematician turned AI educator who explains **ML and deep learning through visual intuition and storytelling** - making hard math approachable without losing the rigor.

Visual

Intuitive, story-driven teaching

Math

Makes ML math genuinely clear

Book

Author of Grokking ML

KNOWN FOR

Grokking ML book and visual explainers.

WHY FOLLOW

Intuitive ML math, no hand-waving.

BEST FOR

Learners who want to understand the why.

FIND THEM

serrano.academy · @SerranoAcademy

AI RESEARCHER · SGD PIONEER

Léon Bottou

Meta FAIR · SGD · leon.bottou.org

The researcher who pioneered **stochastic gradient descent at scale** - the optimization foundation that every deep learning model, from ResNet to GPT-4, is trained with.

SGD
Pioneered large-scale stochastic GD

Meta FAIR
Research scientist at Meta AI

Cited
Among the most cited ML researchers

KNOWN FOR

SGD, large-scale learning, PyTorch roots.

WHY FOLLOW

Fundamental optimization theory in practice.

NOW AT

Meta FAIR research scientist.

FIND THEM

leon.bottou.org

The L stack, in one card.

Save it. Drop a comment with the L-term you use most.

01	Lemmatization	NLP
02	Learning Rate	training
03	LSTM	architecture
04	Linear Regression	learning
05	LoRA	fine-tuning
06	Logistic Regression	learning
07	LLMs	models
08	Loss Function	training
09	Label Encoding	data
10	LR Scheduler	training
11	Layer Normalization	architecture
+	LangChain · LlamaIndex · Llama	tools
+	LM Arena · LM Studio · Labelbox	tools
▶	Latent Space	channel
@	Laurence · Luis · Léon	people

FOLLOW FOR THE FULL A TO Z

Prasanna Kulkarni

in/prasanna-kulkarni-032650180

Next